FACULTY OF COMPUTING

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SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 2

TECHNICAL REPORT

TOPIC: DATA QUALITY IN DATA ENGINEERING

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DATA QUALITY IN DATA ENGINEERING

Executive Summary

In modern data engineering, data quality is considered a crucial factor. The importance of transforming static data attributes into dynamic process behaviours is emphasised. Therefore, real-time analysis and artificial intelligence tools and technologies are being adopted by more and more organisations. Problems related to the scale, speed, and capacity of modern big data can no longer be solved by traditional, manual, and passive troubleshooting methods.

It is shown by the evaluation results that features such as automatic verification, conversion strategy, and pattern execution can be directly embedded into the data ingestion and storage links. In this way, data quality can be improved. Tools such as Great Expectation and Apache Deequ are designed to continuously monitor key features. At present, correctness, consistency, and similar characteristics are considered very important to be monitored.

A proactive strategy is adopted in the report, in which data is treated as a product, and a reliable data lake is aimed to be built. The main strategic suggestions include that all data should be modelled at the design stage and that metadata support should be provided for governance work. A comprehensive strategy can be used to ensure that a reliable basis for enterprise decision-making is provided by downstream business intelligence and machine learning.

1.0 Introduction

Data is only useful when people can trust it. In the data engineering field, data quality simply means the data is good enough to do its thing The data must be accurate, complete, consistent, and on time when it is flowing through the pipeline. When a problem happens with any link in the supply chain, all downstream reports, ML models, and business tools will get wrong inputs. This results in poor judgments.

This holds big significance for businesses. Inventory systems, credit score models, and artificial intelligence require quality data to work efficiently. According to a survey published in 2024, nearly all big companies fight data quality issues. According to a study done by Anomalo and Dimensional Research, most companies state that they can't rely on rule-based checks anymore. It has a price. Every year companies waste millions of dollars on bad data. Meanwhile, solutions that study risks in company's business sectors clearly point out the rapid inflation that is currently underway.

Data pipelines are very difficult today. Data shuffles between on-premise servers, various clouds, and external APIs. Any link could snap. Since companies rely on real-time analysis engineers have to spot problems immediately not hours later. As businesses invest increasingly in machine learning and generative artificial intelligence, the risk of garbage in, garbage out is now greater than ever.

The data engineering aspect of data quality is reviewed in this article. It emphasizes the pipelines, logic, and tools that transfer data between the source and the user. We won't discuss how systems collect the data in the first place or how analysts, in the end, use the data. Alternatively, this paper will reveal how engineers define quality, where pipelines typically fail, and what techniques organizations use today. Automated testing, data observability, data contracts are some methods.

2.0 Background / Literature Review

The quality of data has been studied for decades by scientists. Wang and Strong (1996) were among the first researchers to define it from the point of view of users. Applying it, In simpler words, Data quality is determined by whether or not the data actually serves to help people to complete their task. They identified four components: accuracy, integrity, consistency, and time. Even today, these four components are commonly used by data teams.

However, those four parts are not sufficient anymore. The classic elements fall short of some critical details in the world of big data, Ridzuan and their team (2024) show. In this day and age, data is being generated at incredible speeds. In order to solve the problem, a new list was made with 20 parts, more suited to the complexity of data.

Word meanings cause another problem. Ehrlinger and Wöß have assessed 13 data quality tools and found big differences. The first tool that checks if the row exists calls that "integrity" while the other tool that uses exactly the same word says column has an empty space. When team members use tools with different meanings it is impossible to compare results and agree on quality rules. The absence of any mutual language creates a significant void in the field.

In addition to these definitions, the researchers look at how data pipelines behave. According to a study conducted by Foidl et al. (2024), there are Developer Communities for GitHub and Stack overflow as developer groups. 33% data issues triggered by misguided data types, they discovered. Most of these problems occur when the developers clean the data. Integrating and combining data is the cause of nearly half of developer issues. What this demonstrates is that the quality problems may often start inside the data pipeline and not just in the starting data.

In the business world, companies focus on two main ways to fix these issues. One of the methods is data observability wherein teams use machine learning to monitor data pipelines constantly to flag any unusual patterns. For example, if a table normally gets 500,000 rows but only gets 12,000 rows, the system might flag it. This goes beyond just a one-time check. A data contract is a specified agreement between the people that make the data and the people that use it. The agreement details the precise shape and quantity the data must possess. This approach is inspired by the software engineering idea of finding errors early. The contract prevents any bad changes occuring after the data reaches downstream systems (Metaplane, 2023).

Major challenges remain - even with these tools. Overall, teams are still missing common ways to measure quality across systems (Ehrlinger and Wöß, 2022). It is more difficult for engineers to ensure good quality through spread-out setups like data grids. Ultimately, when it comes to measuring data quality well when creating generative artificial intelligence, researchers still do not know how (Ridzuan et al., 2024; Foidl et al., 2024).

3.0 Analysis and Discussion

In modern data engineering pipelines, data quality is implemented through series of validation, transformation and monitoring processes that embedded across different stages of data processing. Quality controls are integrated into data ingestion, transformation and storage processes instead of being applied on final step as to ensure that errors are detected early and do not propagate to the downstream system (Achanta & Boina, 2023).

3.1 Data Quality Works in data Engineering Pipeline

At the data ingestion stage, validation rules including missing values, incorrect data formats and constraints violations checked are applied to filter out the invalid or incomplete data before it enters the system. During data transformation stage, data cleaning techniques such as deduplication, normalization and standardization are implemented to improve the consistency and usability across datasets. In the stage of storage, schema enforcement ensures that data follows to the predefined structures. It helps in reducing inconsistencies when data is accessed for analysis and modelling. Moreover, modern data engineering adopts to automated tools like Great Expectations and Apache Deequ as these tools allow engineers to define data quality rules and monitor datasets continuously (Kumar & Singh, 2026).

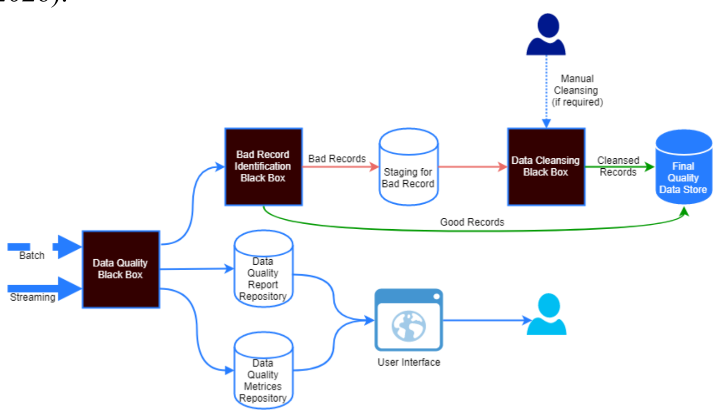

Figure 1 Data Quality Validation and Cleansing Pipeline using Amazon Deequ and Apache Spark (Source: Chakraborty, n.d.)

As shown in Figure 1, data quality processes can be integrated into the data pipeline to validate incoming data, separate bad records and allow only clean data proceeds to downstream systems. The process is supported further through data observability which allow real-time tracking of any data anomalies like sudden changes in volume, schema, and distribution.

3.2 Real World Application of Data Quality

These mechanisms are widely applied in real-world systems. In e-commerce platform, data quality controls prevent duplicate orders, customer’s data missing, wrong pricing and others that would impact user experience. In financial systems, high data quality is a crucial strategic asset as it directly affects the risk management, operational efficiency and decision-making. For instance, an inaccurate transaction record in banking system may propagate through processing and storage stages, this may bring inconsistent account balances and unreliable financial reports. The quality of data has a direct influence on the operational efficiency and business performance.

3.3 Advantages and Disadvantages of Data Quality Implementation

The implementation of data quality frameworks brings advantages. It enhances the reliability of analytics outputs and reduces the need for repeated data cleaning, thereby increasing the organization’s confidence in decision-making, cost and time saving. High quality data ensures accuracy and consistency, thereby enhancing the efficiency of downstream applications such as machine learning models and analytics platforms (Kumar & Singh, 2026). However, these advantages have certain drawbacks. Designing and managing data quality system is high complex as it requires specialized expertise. Additionally, enforcing strict validation rules may introduce latency, especially in today’s fast data processing environment as each data record must undergo comprehensive validation which leads to increasing processing overhead.

3.4 Challenges of Data Quality Implementation

In big data environments, maintaining data quality is more challenging. Data heterogeneity, which refers to the data comes from diverse sources in structured, semi-structured and unstructured format, making data integration and standardization difficult. Missing data is another major challenge as it requires different handling strategies to avoid multiple types of bias such as Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) (Gudivada et al., 2017). Additionally, data quality decay may occur as data quality degrade over time due to system updates, transformations and evolving business requirements.

3.5 Future Trends of Data Quality Implementation in Data Engineering

In the future, data quality management will become more automatic and scalable. Machine learning techniques are now increasingly combined with rule-based validation system to improve anomalies detection and data correction processes. Simultaneously, data observability tools are the new vital for monitoring pipelines on an ongoing basis. The implementation of DataOps practices is enhancing the collaboration between data engineers and stakeholders, thereby allowing faster identification and resolution of data issues (Rahul Cherekar, 2020). As organizations increasingly rely on data-driven decisions, the demand for effective, automated, and real-time data quality solutions will continue to rise.

4.0 Recommendations

Quality control is important in modern data engineering. Engineers should not wait until problems arise before taking measures; instead, they should adopt automated verification techniques from the very beginning of data collection and processing. This is because data processing involves high risks and poses significant threats to the decision-making process, quality must be regarded as an essential part of the entire data lifecycle.These controls should be implemented continuously to ensure that the enterprise has highly reliable systems (Achatana and Boyna, 2023; Reis and Hosley, 2022).

Beyond automation, teams need to be actively involved in observing critical quality dimensions such as accuracy, completeness, consistency, and timeliness (Jawadwani et al., 2015; Sidi et al., 2012). Particularly important is the specific challenges that arise when applying standard tracking strategies to big data infrastructures (Batini and Scannapicco, 2016) due to their sheer scale and speed (Gao et al., 2016). Therefore, quality initiatives should begin during the design phase. This avoids anomalies by default and enforces meaningful business concepts through proactive data modeling before data collection (Hoberman, 2023).

Finally, data governance is crucial for maintaining a complex enterprise ecosystem. To help practitioners quickly identify the causes of data quality degradation, metadata and data analytics tools are essential. By tracking the data path, successfully resolving abnormal situations and ensuring the security of downstream business intelligence and machine learning products, data engineers can identify the causes of faults. Ultimately, a stable and reliable database environment can be enabled by all these measures.

5.0 Conclusion

In conclusion, the key task of enterprise system analysis is considered to be the management of data quality. Nowadays, the speed of data control by companies is increasing. In this case, those old verification technologies have been gradually eliminated, and more attention needs to be paid to automation and "shift-left" engineering. The data flow entity should be extracted, and it should be checked whether the observability framework and verification rules are still valid. This is by no means a one-time activity, but should be carried out regularly so that the best data quality can be ensured.

When data is regarded as a product, it can be ensured that downstream machine learning models and business intelligence tools can be run normally and reliably. As time goes by, the data architecture is made more and more complex, and the number of artificial intelligence applications in quality management and fault prediction is also increased. To make reliable data-based decisions, good data quality is required, and quality assurance work must be carried out before data is generated.

6.0 References

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7.0 Appendix/Logbook

No	Purpose	AI Tool Used	Prompt	Summary of AI response	Verification Steps
1	Understand basic concept	ChatGPT	"What is data quality in data engineering?"	Explained definition, importance, and key dimensions (accuracy, completeness, consistency)	Compared with textbook and academic sources
2	Get real-world examples	ChatGPT	"Give me real-world examples of data quality used in different"	Provided examples in banking, healthcare, e-commerce	Cross-checked with online academic sources (Google Scholar)
3	Understand how data quality applies in data pipeline	ChatGPT	"How is data quality applied in data engineering? Provide the steps and example"	Described data quality implementation across pipeline stages including ingestion, transformation, and storage with validation, cleaning, and schema enforcement examples	Cross-referenced with academic papers and supporting literature and verified with research paper (Achanta & Boina, 2023)

4	Understand the advantages and disadvantages of data quality	ChatGPT	“Provide advantages and disadvantages of data quality with reference support”	Explained benefits such as improved decision-making, reliability, and efficiency, as well as drawbacks like complexity, cost, and processing latency	Cross-referenced with academic papers and supporting literature
5	Understand the challenge and future trends of data quality	ChatGPT	“Provide the challenge faced in applying data quality and future trends with papers and reference support”	Identified challenges such as data heterogeneity, missing data, and data quality decay, and discussed future trends including automation, machine learning integration, and DataOps practices	Cross-referenced with academic papers and supporting literature, verified using Gudivada et al. (2017) and Cherekar (2020)
6	Differentiate Apache Spark and Apache Deequ	ChatGPT	“Provide the function differences between Apache Spark and Apache Deequ with usage examples”	Explained that Apache Spark is a data processing engine, while Apache Deequ is a data quality validation library built on top of Spark, with examples of validation use cases	Confirmed with official documentation and technical articles
7	Refine the structure of strategic recommendations	Google Gemini	“What are the best practices for data quality frameworks in industry?	Offered strategic suggestions on shift-left validation, data observability, and the mindset of data as a product. .	Verified with research paper- *Fundamentals of Data Engineering* (Reis & Housley, 2022).
8	Differentiate executive summary and conclusion part	Google Gemini	"What are the key components of a technical executive summary and conclusion for	Provided the difference between how to write executive summary and conclusion	Compared with the assignment template and rubric to ensure proper word counts

			academic writing report?"		and section focus.
9	Clarify technical concepts	Google Gemini	“What means Garbage in Garbage out principle?”	Elaborated how the quality of output depends on the quality of the input and the prevention mechanisms such as validation at an early stage.	Linked with the Introduction section to maintain the conceptual consistency.
10	Refine technical writing tone	Google Gemini	"How to write a technical report to sound more professional?"	Recommended to use more powerful technical verbs and improving sentence flow for better readability.	Checked against the Technical Writing Rubric for "Writing Quality" marks.

Table 1 AI Tools Used

Detailed report

23/03/2026 - 16/04/2026

Total: 28:18:33   Billable: 28:18:33   Amount: 0,00 USD

clockify

Date	Description	Duration	User
23/03/2026	Distribute task to members Topic 15 - Distribute Task	00:25:19 21:06:00 - 21:31:19	Poh Lok Yee 0,00 USD
25/03/2026	Explore and understand data quality's knowledge Topic 15 - Explore Data Quality Topic	00:37:07 12:39:00 - 13:16:07	Poh Lok Yee 0,00 USD
26/03/2026	Research about data quality topic Topic 15 - Explore Data Quality Topic	00:46:09 13:17:37 - 14:03:46	wuchenxi202603 0,00 USD
27/03/2026	Research and write the introduction part Topic 15 - Research & Write Introduction	01:25:41 13:18:00 - 14:43:41	wuchenxi202603 0,00 USD
28/03/2026	research and write about background literature Topic 15 - Research & Write Background / Literature Review	02:55:10 16:12:00 - 19:07:10	wuchenxi202603 0,00 USD
28/03/2026	Researched the core concepts of the Data Quality topic Topic 15 - Explore Data Quality Topic	01:00:00 20:00:00 - 21:00:00	lauyeewen 0,00 USD
28/03/2026	Put reference inside list and update logbook in clockify Topic 15 - Final Adjustment: Reference + Logbook	00:08:10 20:44:00 - 20:52:10	wuchenxi202603 0,00 USD
29/03/2026	Research and write about how data quality works in data engineering Topic 15 - Write Analysis and Discussion: Data Pipeline Integration	01:43:42 15:12:00 - 16:55:42	Poh Lok Yee 0,00 USD
29/03/2026	reseach and write about case example of data quality in data engineering Topic 15 - Write Analysis and Discussion: Case Examples	01:37:00 20:08:00 - 21:45:00	Poh Lok Yee 0,00 USD
31/03/2026	Research and write about advantages and disadvantages Topic 15 - Write Analysis and Discussion: Advantages and Disadvantages	01:42:31 20:56:00 - 22:38:31	Poh Lok Yee 0,00 USD
01/04/2026	Developed strategic recommendations based on the findings Topic 15 - Write Recommendations	02:00:00 21:00:00 - 23:00:00	lauyeewen 0,00 USD
03/04/2026	Research and write about challenges of data quality Topic 15 - Write Analysis and Discussion: Challenges and Future Trends in Data Quality	01:54:24 15:18:00 - 17:12:24	Poh Lok Yee 0,00 USD
04/04/2026	Research and write about the future trends of data quality Topic 15 - Write Analysis and Discussion: Challenges and Future Trends in Data Quality	01:31:07 20:34:00 - 22:05:07	Poh Lok Yee 0,00 USD
05/04/2026	Summarized the final results and key takeaways Topic 15 - Write Conclusion	02:00:00 20:00:00 - 22:00:00	lauyeewen 0,00 USD

Assignment 1 - Academic Writing   Created with Clockify   1

Figure 2 Clockify Logbook Group 1

06/04/2026	Put reference, update and check clockify Topic 15 - Final Adjustment: Reference + Logbook	00:05:21 00:38:00 - 00:43:21	Poh Lok Yee 0,00 USD
06/04/2026	Created a short overview for a quick recap Topic 15 - Write Executive Summary	02:00:00 20:00:00 - 22:00:00	lauyeewen 0,00 USD
08/04/2026	Finalized the Logbook and formatted all technical references to ensure academic compliance Topic 15 - Final Adjustment: Reference + Logbook	01:00:00 20:00:00 - 21:00:00	lauyeewen 0,00 USD
09/04/2026	Do presentation slide Topic 15 - Presentation slides	00:38:56 11:10:00 - 11:48:56	wuchenxi202603 0,00 USD
09/04/2026	Do slides in canva Topic 15 - Presentation slides	00:47:07 23:24:00 - 00:11:07 +1	Poh Lok Yee 0,00 USD
10/04/2026	Created slides and structured the presentation flow Topic 15 - Presentation slides	01:30:00 13:30:00 - 15:00:00	lauyeewen 0,00 USD
11/04/2026	Conducted a full review and rehearsal to ensure readiness for the live session Topic 15 - Prepare for presentation	01:30:00 12:00:00 - 13:30:00	lauyeewen 0,00 USD
11/04/2026	Prepare script for presentation Topic 15 - Prepare for presentation	00:36:14 12:47:00 - 13:23:14	Poh Lok Yee 0,00 USD
11/04/2026	Prepare script for presentation Topic 15 - Prepare for presentation	00:24:35 13:18:00 - 13:42:35	wuchenxi202603 0,00 USD

Figure 3 Clockify Logbook Group 1